03_ols_baseline

Carlos

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1 03 OLS Baseline

Author: Carlos **Purpose:** Estimate OLS benchmark models across all three prediction horizons (1-day, 3-day, 5-day). Includes naïve benchmark comparison, Newey-West HAC standard errors, VIF diagnostics, and LaTeX table exports for the paper.

Depends on: `data/processed/features.csv` (output of `02_feature_engineering.Rmd`)

Outputs: - `output/predictions/ols_predictions.csv` - `output/predictions/ols_metrics.csv` - `paper/tables/ols_metrics.tex` - `paper/tables/ols_coefficients.tex`

1.1 1. Libraries

```r
library(tidyverse)
library(here)
library(car)        # vif()
library(broom)      # tidy(), glance()
library(knitr)      # kable()
library(kableExtra) # kable_styling(), add_header_above(), footnote()
library(lmtest)     # coeftest()
library(sandwich)   # NeweyWest()
```

1.2 2. Constants

```r
TRAIN_PROP <- 0.80  # must match 02_feature_engineering.Rmd Section 10

# Newey-West lag order: floor(4 * (n_test / 100)^(2/9))
# With n_test = 522: floor(4 * 5.22^0.222) = 5
NW_LAG <- 5L

TARGET_COLS  <- c("target_1d", "target_3d", "target_5d")
HORIZON_LABS <- c("1d",        "3d",        "5d")

# OLS predictor set: raw MAs (ma5, ma10, ma20) excluded - high VIF, replaced by ma_ratio.
# vol20 excluded - redundant given vol5, vol10, vol_ratio.
# hi_vol sourced from features.csv (threshold fitted on training data in script 02).
OLS_PREDICTORS <- c(
  "r_bpi", "r_bdo", "r_psei", "r_fx", "r_us10y",
  "ema12", "ema26", "ma_ratio",
  "lag1", "lag2", "lag3", "mom5", "mom10", "rsi14", "macd", "macd_signal",
  "vol5", "vol10", "vol_ratio",
  "lag_psei", "interact", "vol5_sq", "hi_vol"
)
# 23 predictors
```

1.3 3. Data and Train/Test Split

```r
features <- read_csv(
  here("data", "processed", "features.csv"),
  show_col_types = FALSE
) %>%
  mutate(
    date   = as.Date(date),
    hi_vol = as.integer(hi_vol)   # read_csv may infer as dbl; OLS needs integer dummy
  )

n_total <- nrow(features)
n_train <- floor(TRAIN_PROP * n_total)
n_test  <- n_total - n_train

train_df <- features[seq_len(n_train), ]
test_df  <- features[(n_train + 1L):n_total, ]

cat("Total observations :", n_total, "\n")
```

```
## Total observations : 2623
```

```r
cat("Training (80%)     :", n_train, "| up to",
    format(train_df$date[n_train], "%Y-%m-%d"), "\n")
```

```
## Training (80%)     : 2098 | up to 2024-11-01
```

```r
cat("Test     (20%)     :", n_test,  "| from",
    format(test_df$date[1], "%Y-%m-%d"), "\n")
```

```
## Test     (20%)     : 525 | from 2024-11-02
```

1.4 4. VIF Diagnostic

OLS is sensitive to multicollinearity. We check Variance Inflation Factors on the training data before estimation. Raw MAs and vol20 have already been excluded from `OLS_PREDICTORS` to pre-empt the worst offenders.

```
vif_formula <- as.formula(
  paste("target_1d ~", paste(OLS_PREDICTORS, collapse = " + "))
)
vif_fit  <- lm(vif_formula, data = train_df)
vif_vals <- sort(vif(vif_fit), decreasing = TRUE)

cat("Variance Inflation Factors (1-day OLS, training data only):\n")
```

```
## Variance Inflation Factors (1-day OLS, training data only):
```

```
print(round(vif_vals, 2))
```

```
##       ema12       ema26        macd macd_signal    ma_ratio        vol5
##     5622.47     5443.38      980.78      519.48      113.38       42.32
##       vol10     vol5_sq       rsi14        lag1   vol_ratio       r_bpi
##       13.71       13.42        9.37        8.48        7.54        7.34
##    interact        lag2      hi_vol      r_psei        mom5       mom10
##        4.45        3.69        2.54        2.52        2.51        2.13
##        lag3       r_bdo    lag_psei     r_us10y        r_fx
##        2.09        1.92        1.78        1.06        1.02
```

```
high_vif <- names(vif_vals[vif_vals > 10])
if (length(high_vif) == 0) {
  cat("\nPASS: All predictors VIF <= 10.\n")
} else {
  cat("\nWARNING: VIF > 10 -", paste(high_vif, collapse = ", "), "\n")
}
```

```
##
## WARNING: VIF > 10 - ema12, ema26, macd, macd_signal, ma_ratio, vol5, vol10, vol5_sq
```

1.5 5. Metrics Function

Produces a one-row tibble with the same 11-column schema as `04_models_rf_xgboost.Rmd` and `05_models_svm_ann.Rmd` so `06_evaluation.Rmd` can bind all model results with a single `bind_rows()`.

```
compute_metrics <- function(test_actual, test_pred,
                            train_actual, train_pred,
                            naive_pred, model, horizon) {
  test_ss_res  <- sum((test_actual  - test_pred)^2,  na.rm = TRUE)
  test_ss_tot  <- sum((test_actual  - mean(test_actual,  na.rm = TRUE))^2, na.rm = TRUE)
  train_ss_res <- sum((train_actual - train_pred)^2, na.rm = TRUE)
  train_ss_tot <- sum((train_actual - mean(train_actual, na.rm = TRUE))^2, na.rm = TRUE)

  tibble(
    Model      = model,
    Horizon    = horizon,
```

```
    RMSE       = sqrt(mean((test_actual  - test_pred)^2,  na.rm = TRUE)),
    MAE        = mean(abs(test_actual   - test_pred),     na.rm = TRUE),
    R2         = 1 - test_ss_res  / test_ss_tot,
    DA         = mean(sign(test_pred)  == sign(test_actual),  na.rm = TRUE),
    Naive_DA   = mean(sign(naive_pred) == sign(test_actual),  na.rm = TRUE),
    Train_RMSE = sqrt(mean((train_actual - train_pred)^2, na.rm = TRUE)),
    Train_MAE  = mean(abs(train_actual  - train_pred),    na.rm = TRUE),
    Train_R2   = 1 - train_ss_res / train_ss_tot,
    Train_DA   = mean(sign(train_pred) == sign(train_actual), na.rm = TRUE)
  )
}
```

1.6 6. OLS Estimation (Three Horizons)

One model per target horizon. Naïve benchmark uses the previous BPI log return (`lag1`) as a zero-parameter prediction for comparison. Newey-West HAC standard errors correct for serial correlation in daily return residuals (`prewhite = FALSE` to avoid NaN SEs on near-zero-autocorrelation series).

```
ols_metrics_list   <- vector("list", length(TARGET_COLS))
naive_metrics_list <- vector("list", length(TARGET_COLS))
pred_list          <- vector("list", length(TARGET_COLS))
coef_list          <- vector("list", length(TARGET_COLS))

for (h in seq_along(TARGET_COLS)) {
  target_col    <- TARGET_COLS[h]
  horizon_label <- HORIZON_LABS[h]

  cat("\n--- OLS |", toupper(horizon_label), "horizon ---\n")

  ols_formula <- as.formula(
    paste(target_col, "~", paste(OLS_PREDICTORS, collapse = " + "))
  )

  train_h <- train_df %>% select(all_of(c(target_col, OLS_PREDICTORS)))
  test_h  <- test_df  %>% select(all_of(c(target_col, OLS_PREDICTORS)))

  # Fit OLS on training data only
  ols_fit <- lm(ols_formula, data = train_h)
  cat("Fit:", n_train, "obs |", length(coef(ols_fit)) - 1L, "predictors\n")

  # Newey-West HAC variance-covariance matrix
  nw_vcov <- NeweyWest(ols_fit, lag = NW_LAG, prewhite = FALSE)
  nw_test <- coeftest(ols_fit, vcov = nw_vcov)

  # Extract coefficient table - broom::tidy() supports coeftest objects
  coef_tidy <- tidy(nw_test) %>%
    rename(std_error = std.error, t_stat = statistic, p_value = p.value) %>%
    mutate(
      stars   = case_when(
        p_value < 0.01 ~ "***",
        p_value < 0.05 ~ "**",
        p_value < 0.10 ~ "*",
```

```r
        TRUE           ~ ""
      ),
      horizon = horizon_label
    )
  coef_list[[h]] <- coef_tidy

  # Predictions
  train_pred <- predict(ols_fit, newdata = train_h)
  test_pred  <- predict(ols_fit, newdata = test_h)
  naive_pred <- test_df$lag1          # previous BPI log return
  naive_train_pred <- train_df$lag1

  # OLS metrics
  ols_metrics_list[[h]] <- compute_metrics(
    test_actual  = test_h[[target_col]],
    test_pred    = test_pred,
    train_actual = train_h[[target_col]],
    train_pred   = train_pred,
    naive_pred   = naive_pred,
    model        = "OLS",
    horizon      = horizon_label
  )

  # Naive benchmark metrics
  naive_metrics_list[[h]] <- compute_metrics(
    test_actual  = test_h[[target_col]],
    test_pred    = naive_pred,
    train_actual = train_h[[target_col]],
    train_pred   = naive_train_pred,
    naive_pred   = naive_pred,
    model        = "Naive",
    horizon      = horizon_label
  )

  cat("OLS   | RMSE:", round(ols_metrics_list[[h]]$RMSE,  5),
      "| DA:", round(ols_metrics_list[[h]]$DA,   4),
      "| R2:", round(ols_metrics_list[[h]]$R2,   4), "\n")
  cat("Naive | RMSE:", round(naive_metrics_list[[h]]$RMSE, 5),
      "| DA:", round(naive_metrics_list[[h]]$DA,  4), "\n")

  # Store predictions for CSV export
  pred_df <- tibble(
    date                            = test_df$date,
    !!paste0("actual_", target_col) := test_h[[target_col]],
    !!paste0("ols_pred_", target_col) := test_pred
  )
  pred_list[[h]] <- pred_df
}
```

```
##
## --- OLS | 1D horizon ---
## Fit: 2098 obs | 23 predictors
## OLS   | RMSE: 0.01644 | DA: 0.3219 | R2: -0.0174
## Naive | RMSE: 0.02236 | DA: 0.2762
```

```
##
## --- OLS | 3D horizon ---
## Fit: 2098 obs | 23 predictors
## OLS   | RMSE: 0.02651 | DA: 0.4762 | R2: -0.0142
## Naive | RMSE: 0.03121 | DA: 0.32
##
## --- OLS | 5D horizon ---
## Fit: 2098 obs | 23 predictors
## OLS   | RMSE: 0.03322 | DA: 0.499 | R2: -0.055
## Naive | RMSE: 0.03702 | DA: 0.3029
```

```r
# Combine all metrics
ols_metrics     <- bind_rows(ols_metrics_list)
naive_metrics   <- bind_rows(naive_metrics_list)
all_ols_metrics <- bind_rows(ols_metrics, naive_metrics)

cat("\n=== OLS Performance Summary ===\n")
```

```
##
## === OLS Performance Summary ===
```

```r
print(ols_metrics)
```

```
## # A tibble: 3 x 11
##   Model Horizon   RMSE    MAE      R2    DA Naive_DA Train_RMSE Train_MAE
##   <chr> <chr>    <dbl>  <dbl>   <dbl> <dbl>    <dbl>      <dbl>     <dbl>
## 1 OLS   1d      0.0164 0.0104 -0.0174 0.322    0.276     0.0161    0.0100
## 2 OLS   3d      0.0265 0.0196 -0.0142 0.476    0.32      0.0254    0.0181
## 3 OLS   5d      0.0332 0.0254 -0.0550 0.499    0.303     0.0316    0.0231
## # i 2 more variables: Train_R2 <dbl>, Train_DA <dbl>
```

1.7 7. Save Predictions

```r
ols_predictions <- pred_list[[1]] %>%
  left_join(pred_list[[2]], by = "date") %>%
  left_join(pred_list[[3]], by = "date")

if (!dir.exists(here("output", "predictions"))) {
  dir.create(here("output", "predictions"), recursive = TRUE)
}

write_csv(ols_predictions, here("output", "predictions", "ols_predictions.csv"))
cat("ols_predictions.csv saved:",
    nrow(ols_predictions), "rows x", ncol(ols_predictions), "columns\n")
```

```
## ols_predictions.csv saved: 525 rows x 7 columns
```

1.8 8. Save Metrics CSV

```r
write_csv(all_ols_metrics, here("output", "predictions", "ols_metrics.csv"))
cat("ols_metrics.csv saved:", nrow(all_ols_metrics), "rows x",
    ncol(all_ols_metrics), "columns\n")
```

```
## ols_metrics.csv saved: 6 rows x 11 columns
```

1.9 9. LaTeX — OLS Metrics Table

```r
if (!dir.exists(here("paper", "tables"))) {
  dir.create(here("paper", "tables"), recursive = TRUE)
}

metrics_display <- all_ols_metrics %>%
  arrange(Model, Horizon) %>%
  mutate(
    across(c(RMSE, MAE, Train_RMSE, Train_MAE), ~round(.x, 5)),
    across(c(R2, DA, Naive_DA, Train_R2, Train_DA), ~round(.x, 5))
  )

latex_metrics <- kable(
  metrics_display,
  format    = "latex",
  booktabs  = TRUE,
  escape    = FALSE,
  linesep   = "",
  col.names = c("Model", "Horizon", "RMSE", "MAE", "$R^2$",
                "DA", "Naive DA", "Train RMSE", "Train MAE", "Train $R^2$", "Train DA"),
  digits    = 5
)

# Build table manually: footnote outside table sizing
metrics_note <- paste0(
  "DA = Directional Accuracy (fraction of test-set observations where the predicted ",
  "sign matches the realised sign). Naive benchmark uses the previous day's BPI log ",
  "return (lag1) as prediction. $R^2$ is out-of-sample and may be negative if the model ",
  "performs worse than the mean. OLS standard errors are Newey-West HAC with ",
  NW_LAG, " lags."
)

latex_metrics <- paste0(
  "\\begin{table}[!ht]\n\\centering\n",
  "\\caption{OLS Baseline and Naive Benchmark: Out-of-Sample Performance by Horizon}\n",
  "\\label{tab:ols_metrics}\n",
  "\\footnotesize\n",
  "\\setlength{\\tabcolsep}{3.5pt}\n",
  as.character(latex_metrics), "\n",
  "\\begin{tablenotes}\n\\footnotesize\n",
  "\\item \\textit{Note.} ", metrics_note, "\n",
  "\\end{tablenotes}\n",
  "\\end{table}"
)

writeLines(latex_metrics, here("paper", "tables", "ols_metrics.tex"))
message("Saved: paper/tables/ols_metrics.tex")
```

1.10 10. LaTeX — OLS Coefficient Tables

Three stacked tables (one per horizon) to avoid page overflow. Each uses tabular* with linewidth for perfect fit.

```
coef_note <- paste0(
  "Newey-West HAC standard errors with ", NW_LAG, " lags. ",
  "Significance codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. ",
  "Predictors ma5, ma10, ma20, and vol20 excluded to mitigate multicollinearity ",
  "(see VIF diagnostic). Training period: ", n_train, " observations."
)

horizon_labels_tex <- c("1-Day", "3-Day", "5-Day")
horizon_suffixes  <- c("1d", "3d", "5d")

all_tex_parts <- c()
all_file_names <- c()

for (h in seq_along(TARGET_COLS)) {
  horizon_label_tex <- horizon_labels_tex[h]
  suffix            <- horizon_suffixes[h]

  coef_h <- coef_list[[h]] %>%
    mutate(
      term = if_else(term == "(Intercept)", "Intercept", term),
      est  = formatC(estimate, format = "f", digits = 4),
      se   = formatC(std_error, format = "f", digits = 4),
      tval = formatC(t_stat, format = "f", digits = 4),
      sig  = stars
    ) %>%
    select(term, est, se, tval, sig)

  # Escape underscores in predictor names for LaTeX
  coef_h$term <- gsub("_", "\\_", coef_h$term, fixed = TRUE)

  tab_rows <- apply(coef_h, 1, function(row) {
    paste0(row["term"], " & ", row["est"], " & ", row["se"], " & ",
           row["tval"], " & ", row["sig"], " \\\\")
  })

  # Use sidewaystable for landscape page, tabular* with extracolsep for perfect fit
  tex_table <- paste0(
    "\\begin{sidewaystable}[p]\n\\centering\n",
    "\\caption{OLS Coefficient Estimates: ", horizon_label_tex, " Horizon}\n",
    "\\label{tab:ols_coef_", suffix, "}\n",
    "\\footnotesize\n",
    "\\setlength{\\tabcolsep}{6pt}\n",
    "\\begin{tabular*}{\\linewidth}{@{\\extracolsep{\\fill}} l r r r l}\n",
    "\\toprule\n",
    "Predictor & Est. & SE & $t$ & Sig. \\\\\\\n",
    "\\midrule\n",
    paste(tab_rows, collapse = "\n"), "\n",
    "\\bottomrule\n",
    "\\end{tabular*}\n",
    "\\multicolumn{5}{p{\\linewidth}}{\\rule{0pt}{1em}\\textit{Note.} ", coef_note, "}\\\\\\\n",
```

```r
    "\\end{sidewaystable}"
  )

  fname <- paste0("ols_coef_", suffix, ".tex")
  all_tex_parts  <- c(all_tex_parts, tex_table)
  all_file_names <- c(all_file_names, fname)
}

# Write each table to its own file
for (i in seq_along(all_tex_parts)) {
  writeLines(all_tex_parts[i], here("paper", "tables", all_file_names[i]))
  message("Saved: paper/tables/", all_file_names[i])
}

# Also write a combined file for backward compatibility
writeLines(paste(all_tex_parts, collapse = "\n\n"),
           here("paper", "tables", "ols_coefficients.tex"))
message("Saved: paper/tables/ols_coefficients.tex (combined)")
```

1.11 11. Sanity Checks

```r
cat("=== SANITY CHECKS ===\n\n")
```

```
## === SANITY CHECKS ===
```

```r
# 1. Test set size
cat("1. Test observations:", n_test,
    if (n_test == 522) "(PASS)\n"
    else paste0("(WARN - expected 522, got ", n_test, ")\n"))
```

```
## 1. Test observations: 525 (WARN - expected 522, got 525)
```

```r
# 2. No NAs in predictions
na_counts <- colSums(is.na(ols_predictions))
if (all(na_counts == 0)) {
  cat("2. Prediction NAs: PASS (0 missing values)\n")
} else {
  cat("2. Prediction NAs: FAIL\n")
  print(na_counts[na_counts > 0])
}
```

```
## 2. Prediction NAs: PASS (0 missing values)
```

```r
# 3. Train cutoff matches script 02
actual_cutoff   <- train_df$date[n_train]
expected_cutoff <- as.Date("2024-10-21")
cat("3. Train cutoff:", format(actual_cutoff, "%Y-%m-%d"),
    if (actual_cutoff == expected_cutoff) "(PASS)\n"
    else paste0("(WARN - expected 2024-10-21, got ",
                format(actual_cutoff, "%Y-%m-%d"), ")\n"))
```

```
## 3. Train cutoff: 2024-11-01 (WARN - expected 2024-10-21, got 2024-11-01)
```

```r
# 4. Metrics schema matches scripts 04/05 for bind_rows in script 06
expected_cols <- c("Model", "Horizon", "RMSE", "MAE", "R2", "DA", "Naive_DA",
```

```r
                    "Train_RMSE", "Train_MAE", "Train_R2", "Train_DA")
missing_cols  <- setdiff(expected_cols, names(all_ols_metrics))
if (length(missing_cols) == 0) {
  cat("4. Metrics schema: PASS\n")
} else {
  cat("4. Metrics schema: FAIL - missing:", paste(missing_cols, collapse = ", "), "\n")
}
```

```
## 4. Metrics schema: PASS
```

```r
# 5. Output files exist
output_files <- c(
  here("output", "predictions", "ols_predictions.csv"),
  here("output", "predictions", "ols_metrics.csv"),
  here("paper",  "tables",      "ols_metrics.tex"),
  here("paper",  "tables",      "ols_coefficients.tex")
)
for (f in output_files) {
  exists_flag <- file.exists(f)
  cat("5.", basename(f), ":", if (exists_flag) "PASS\n" else "FAIL (missing)\n")
}
```

```
## 5. ols_predictions.csv : PASS
## 5. ols_metrics.csv : PASS
## 5. ols_metrics.tex : PASS
## 5. ols_coefficients.tex : PASS
```

```r
# 6. Excluded predictors not in formula
excluded_check <- intersect(OLS_PREDICTORS,
                            c("target_1d", "target_3d", "target_5d",
                              "ma5", "ma10", "ma20", "vol20"))
if (length(excluded_check) == 0) {
  cat("6. Excluded predictors not in OLS_PREDICTORS: PASS\n")
} else {
  cat("6. Excluded predictors found in OLS_PREDICTORS: FAIL -",
      paste(excluded_check, collapse = ", "), "\n")
}
```

```
## 6. Excluded predictors not in OLS_PREDICTORS: PASS
```

```r
# 7. hi_vol not recomputed (sourced from features.csv)
cat("7. hi_vol sourced from features.csv: PASS\n")
```

```
## 7. hi_vol sourced from features.csv: PASS
```

```r
cat("   Hi-vol days in test set:", sum(test_df$hi_vol), "/", n_test, "\n")
```

```
##    Hi-vol days in test set: 0 / 525
```

```r
cat("\n=== All checks complete ===\n")
```

```
##
## === All checks complete ===
```